Calc 4.3 - Beam B-3, line 3, grids C to D, shear (extract)
By: MK 08/22/2026 Checked: AR 08/27/2026

Basis: ACI 318-19 Chapters 9 and 22. f'c = 4,000 psi (S-001 B1, beams Levels 2 to Roof), normalweight, lambda = 1.0. fyt = 60,000 psi. phi = 0.75 for shear (Table 21.2.1).

Section: bw = 16 in., h = 30 in., d = 27.5 in.

Loads (factored, 1.2D + 1.6L): wu = 4.0 kips/ft including self-weight, plus the girder G-2 reaction of 62 kips at 9'-0" from the face of the grid C column. Column C-3 is 24 in. square, so the face is 1'-0" from the support line. Reaction at grid C = 107.2 kips. Critical section at d from the face of support (9.4.3.2; the support reaction is compressive, loads are applied at the top, and no concentrated load lies between the face and d): **Vu = 94.0 kips**. Grid D: Vu at d = 81.6 kips (does not govern).

Check at d, grid C	Value
Vc = 2 lambda sqrt(f'c) bw d (Table 22.5.5.1(a), Av >= Av,min)	55.7 kips
#4 @ 6 in., 2 legs: Vs = Av fyt d / s = 0.40 x 60 x 27.5 / 6	110.0 kips
phi Vn = 0.75 (55.7 + 110.0)	124.3 kips
Vu / phi Vn	0.76 OK
Max spacing: Vs <= 4 sqrt(f'c) bw d = 111.3 kips, so s,max = d/2 = 13.75 in. (Table 9.7.6.2.2)	OK
Middle zone, #4 @ 12 in.: phi Vn = 0.75 (55.7 + 55.0) = 83.0 kips > Vu at 6'-0" from face = 107.2 - 4.0 x 7.0 = 79.2 kips	OK

Use #4 @ 6 in. within 6'-0" of each support and #4 @ 12 in. in the middle, as scheduled on S-201.